

VAMSEE ACHANTA, P.E.

Project Manager | Marine Terminals, Mooring & Maritime Engineering | Digital Delivery

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Summary

Licensed Professional Engineer (Texas) with 23 years of marine and offshore engineering experience, including more than 10 years leading multidisciplinary engineering teams and managing client projects from scope definition through delivery, including budgets and financial tracking. I have also run business lines and companies with full P&L responsibility as co-founder of FDAS and principal of AceEngineer. My current work is marine terminal engineering: berthing, mooring, fenders and passing-ship analysis for moored vessels, including the \$1.8B WoodFibre LNG terminal. I have managed many concurrent projects: 100+ assignments at 2H Offshore, 12+ years running an independent consulting practice, and the BP Macondo containment riser response, where I was Engineering Manager for a 24/7 team working across several locations. I build Python and AI-assisted workflows that cut analysis time while keeping every result traceable and QA-checked, which fits HDR's push toward digital delivery.

Core Qualifications

- **Project leadership:** scope, schedule, budget and deliverables management; project financial tracking and earned value; business-line and company P&L management; staffing and workload coordination across concurrent projects; client relationship ownership; proposal support and scoping; design reviews and technical reporting.
- **Quality and risk:** QA/QC of analysis deliverables, independent checking, design basis development, contractor deliverable review, technical risk assessment, emergency response.
- **Marine terminals and maritime:** vessel berthing and mooring, shore mooring and fender interaction, passing-ship effects on moored vessels and tugs, operability and weather-window assessment, moored-barge failure investigation, marine operations and installation engineering.
- **Analysis tools:** OrcaFlex, AQWA, OrcaWave, ANSYS, Abaqus; Python and SQL automation; API-driven model generation, batch execution and automated reporting.
- **Standards:** API RP-16Q / 17G / 17G2 (committee work), API 579-1 / ASME FFS-1, BS 7910, DNV recommended practices, ASME VIII.
- **Tools:** MS Project, MS Office (Excel, Word, PowerPoint), Git / Azure DevOps project tracking.

Professional Experience

Naval Architect — Alan McClure & Associates, Houston, TX

Dec 2023 – Present

- **WoodFibre LNG Terminal, British Columbia (\$1.8B):** Led marine global analysis for two Floating Storage Terminals berthed side-by-side in about 35 m water depth. Covered coupled FST and shore-mooring analysis for extreme weather, FST/LNG carrier/fender/mooring interaction for operating conditions, and berthing support for LNG carriers up to 180,000 m³.
- **Passing-ship analysis:** Assessed tug motions and operating risk from passing-ship effects using time-domain analysis. This kind of study applies directly to busy ship channels and berths.
- **Moored-barge failure investigations:** Used diffraction and global analysis to find the root causes of mooring-line failures in extreme weather and recommended improvements to the mooring system.
- **Digital delivery:** Built Python/API workflows for diffraction and mooring analysis that automate model setup, execution, results extraction and report plots. They shorten turnaround and make results repeatable and easy to check.

Engineering Lead Consultant / Principal — AceEngineer (independent practice), Houston, TX

Jun 2012 – Present

- Run a consulting practice serving operators and engineering contractors. I own client relationships, scoping and proposals, budgets, schedules and delivery for concurrent assignments, and manage the business's P&L.
- **ExxonMobil Yellowtail, Guyana:** Umbilical installation analysis in about 6,000 ft of water; organized and automated 100+ load cases and defined installation windows to reduce execution risk.
- **Chevron Ballymore / Talos Venice Limerock:** Installation analysis for rigid jumpers, manifolds and static umbilicals, including installation tables and wave/current operating limits.

- **42-inch subsea pipeline, Venezuela:** FEA of bending capacity for thin-walled pipe outside the standard DNV code range, plus pipelay analysis.
- **Fitness-for-service and integrity:** Python workflows for API 579-1 / BS 7910 assessments and COMSOL modeling of mooring-line corrosion to support inspection-interval recommendations.
- **AI-assisted engineering:** Run an engineering software ecosystem of 25+ repositories with LLM-assisted workflows under automated QA. Every published number traces back to its source data and checked code.

VP of Engineering / Co-Founder — Frontier Deepwater Appraisal Solutions (FDAS), Houston, TX

Jun 2016 – Present

- Co-founded and help run the company: business management, P&L and budgets, and engineering leadership for deepwater development concepts and operator engagement, including technical proposals, cost-benefit studies and business development.
- **Semisubmersible production vessel concept (6,000 ft):** Led the feasibility study for converting idle drillships to production use, including riser support, mooring and hurricane-condition analysis to support investment decisions.
- **SEWOL salvage (Korea):** Delivered emergency hull FEA within 72 hours for government salvage support.
- **Field economics platform:** Built SQL/Python analytics on BSEE data for 200+ Gulf of Mexico fields, cutting field evaluation time from weeks to days.

Engineering SME, Data Science Team — Occidental Petroleum, Houston, TX

Sep 2017 – Dec 2020

- Engineering lead coordinating production engineering, drilling, data science and software teams. Delivered 50+ physics-based algorithms that monitor about 20,000 wells in real time and support 20+ concurrent drilling campaigns.
- Owned requirements, validation and production support for engineering analytics. This is direct experience rolling out digital tools across a large organization.

Engineer → Engineering Lead / Engineering Manager — 2H Offshore Inc, Houston, TX

Aug 2003 – Jun 2015

- Progressed from analyst to lead, manager and technical authority. Managed client communication, technical reports, design reviews, team coordination, mentoring and delivery for 100+ assignments.
- **BP Macondo containment riser (emergency response):** As Engineering Manager, led a 24/7 multidisciplinary team across several locations and delivered a complete containment riser design in 8 weeks by repurposing existing assets.
- **Chevron Bangka riser FEED:** Led FEED for production risers on a shallow-water development and delivered ahead of schedule.
- **Major project portfolio:** BP Thunder Horse, Atlantis and Horn Mountain; Chevron Jack/St. Malo; ENI Devils Tower; Reliance KG-D6; and other projects in the Gulf of Mexico, West Africa and Asia-Pacific.
- **Industry standards:** Committee member and contributor for API RP-16Q and related riser standards.

Education & Licensure

- **Professional Engineer (P.E.)** — Texas
- **M.S., Mechanical Engineering** — Texas A&M University, College Station, TX
- **B.Tech., Mechanical Engineering** — Indian Institute of Technology (IIT) Madras

Selected Publications

- OTC (2017): "Heavy Lift Dynamics — Sewol Ferry Offshore Salvage"
- IADC (2015): Contributing Author, *Floating Drilling Equipment and Operations*, 12th Ed.
- OMAE (2009): "Benchmarking SHEAR7v4.5: Full-Scale Drilling Riser VIV Data"
- World Oil (2024): "Wellbay Innovation Supports Lower-Cost HP DW Discoveries"

Affiliations: ASME | SPE | Marine Technology Society | API Standards Committees